

Q No:8

DS	Session	Searching	Question Information	Level 1	Challenge 8
Problem Description: Ram's first job is a out of university is for a student's book publisher. They are currently publishing a line of "Intelligent libs" books, and they need you to check their print-copy. Write a program that will take a sentence with placeholders in it, and substitute in for those placeholders words from the lists of example words the publisher has provided. Your goal will be to substitute the appropriate words into the sentence from the list of provided words TWICE (keeping track of the words you have used) where: [N] = nouns [AV] = adverbs [V] = verbs [A] = adjectives Input Format: Line one will always be the line with the story. The order of the Nouns, Adverbs, Verbs and Adjectives will always be the same. Input stops when it reaches the word END alone on a line. You are guaranteed to always have enough words to substitute into the sentence TWICE (not repeating any words). The number of words given under each section will be random. Output Format: Print the output in a separate lines contains, Always take the TOP-most UNUSED word from a group when substituting. Output the final sentences when all substitutions are done.					
▼ Logical Test Cases Test Case 1 INPUT (STDIN) There once was a [A] [N] from [N] who [AV] [V] all day. NOUNS squirrel London tree lettuce ADVERBS silently quickly Test Case 2 INPUT (STDIN) There once was a [A] [N] from [N] who [AV] [V] all day. NOUNS singer chennai hotel chettinad ADVERBS silently quickly					
happily VERBS skipped ran climbs ADJECTIVES delightful homely quain EXPECTED OUTPUT There once was a delightful squirrel from London who silently skipped all day. There once was a homely tree from lettuce who quickly ran all day. happily sadly VERBS skipped reads ran sings climbs ADJECTIVES delightful homely surprisingly EXPECTED OUTPUT There once was a delightful singer from chennai who silently skipped all day. There once was a homely hotel from chettinad who quickly reads all day.					
▼ Mandatory Test Cases Test Case 1 KEYWORD int cl[CMDS] Test Case 2 KEYWORD char *lists[CMDS] [MAXWORDS]; Test Case 3 KEYWORD char *tokens[TOKENS]={ "[N]", "[AV]", "[V]", "[A]" }; Test Case 4 KEYWORD char *cmds[CMDS]= { "NOUNS", "ADVERBS", "VERBS", "ADJECTIVE", "S", "END" };					

Code:

```
#include <stdio.h>
#include <string.h>

#define MAX 100
#define SIZE 100
```

```
int main()
{
    char sentence[500];
    char nouns[MAX][SIZE];
    char adverbs[MAX][SIZE];
    char verbs[MAX][SIZE];
    char adjectives[MAX][SIZE];

    int nc = 0, ac = 0, vc = 0, adjc = 0;
    int section = 0;

    char line[SIZE];
    fgets(sentence, sizeof(sentence), stdin);
    sentence[strcspn(sentence, "\n")] = '\0';
    while (fgets(line, sizeof(line), stdin))
    {
        line[strcspn(line, "\n")] = '\0';

        if (strcmp(line, "END") == 0)
            break;

        if (strcmp(line, "NOUNS") == 0)
        {
            section = 1;
            continue;
        }

        if (strcmp(line, "ADVERBS") == 0)
        {
            section = 2;
            continue;
        }

        if (strcmp(line, "VERBS") == 0)
        {
            section = 3;
            continue;
        }

        if (strcmp(line, "ADJECTIVES") == 0)
        {
            section = 4;
            continue;
        }

        if (section == 1)
        {
            strcpy(nouns[nc], line);
```

```
nc++;
}
else if (section == 2)
{
strcpy(adverbs[ac], line);
ac++;
}
else if (section == 3)
{
strcpy(verbs[vc], line);
vc++;
}
else if (section == 4)
{
strcpy(adjectives[adjc], line);
adjc++;
}
}
int ni = 0;
int ai = 0;
int vi = 0;
int adi = 0;

for (int round = 0; round < 2; round++)
{
char temp[500];
strcpy(temp, sentence);

char *token = strtok(temp, " ");
int first = 1;

while (token != NULL)
{
char *word = token;

if (strcmp(token, "[N]") == 0)
{
word = nouns[ni];
ni++;
}
else if (strcmp(token, "[AV]") == 0)
{
word = adverbs[ai];
ai++;
}
else if (strcmp(token, "[V]") == 0)
{
word = verbs[vi];
vi++;
}
else if (strcmp(token, "[A]") == 0)
```

```
{
word = adjectives[adi];
adi++;
}

if (!first)
printf(" ");

printf("%s", word);

first = 0;
token = strtok(NULL, " ");
}

printf("\n");
}

return 0;
}
```

Output:

Test_case 1

```
There once was a [A] [N] from [N] who [AV] [V] all day.
NOUNS
squirrel
London
tree
lettuce
ADVERBS
silently
quickly
VERBS
skipped
ran
ADJECTIVES
delightful
homely
END
There once was a delightful squirrel from London who silently skipped all day.
There once was a homely tree from lettuce who quickly ran all day.
```

Test_case 2

There once was a [AJ] [N] from [N] who [AV] [V] all day.

NOUNS

singer

chennai

hotel

chettinad

ADVERBS

silently

quickly

VERBS

skipped

reads

ADJECTIVES

delightful

homely

END

There once was a [AJ] singer from chennai who silently skipped all day.

There once was a [AJ] hotel from chettinad who quickly reads all day.